

Weekly Progress Report

Refactoring Detection Project - Week 2 progress update for the Monday research supervisor meeting

Based on the Week 2 slide deck and the second meeting notes from 05/25/2026.

1. Overview

During Week 2, the team moved from basic project understanding into testing, UI improvement planning, and project organization. The main focus was running the tool, reviewing logs, identifying useful information for the interface, discussing confidence score ideas, and continuing to organize weekly files through the progress website and shared folders.

2. Monday Meeting Summary - 05/25/2026

- The team presented its understanding of the agents and their functionality.
- The Week 1 presentation was uploaded to the slides folder in the team OneDrive.
- Dhir's prototype was presented as the selected prototype direction.
- Amaro's Summer Research Website was presented as the place to track weekly progress and files.
- The team stopped using the personal group chat and moved communication to the SPARK channel.
- A new private voice channel was created in SPARK for team meetings and discussion.

3. Weekly Progress Summary

Area	Progress Made
Code testing	The team worked on running and testing the project code during the week.
Log review	The team planned to read log files and identify information that could improve the UI design.
Website	The updated website was pushed to a new branch on the GitHub repository created by Dr. AlOmar and is currently hosted online.
Files and documentation	Slides, meeting notes, reports, collaborator files, and other resources are being organized for OneDrive and the website.
UI design	The team gathered ideas for improving the current UI design and making it closer to IntelliJ style.
Confidence score	A dedicated confidence score section was started, with Dhir and Juliana working on this part.
GitHub and communication	The team continued using GitHub branches, shared files, and the SPARK communication channel.

4. Website and Weekly Files

The progress website is being used to keep the research work organized across the weeks. The updated website was pushed to the new GitHub repository created by Dr. AlOmar and is hosted here:

<https://se4airesearch.github.io/clone-summer2026/>.

- Amaro will continue updating the website throughout the weeks.
- The website will include weekly slides, meeting notes, reports, collaborator files, and other documents if needed.
- The same base design from Week 1 files will be reused for consistency.
- Files are also being uploaded and organized in the team OneDrive.

5. Code Testing and Log Review Plan

The team was asked to run the code and test the tool throughout the week. While testing, the team should read the generated log files and look for information that can help improve the UI design.

- Run the project code during the week.
- Test the tool and observe how the workflow behaves.
- Read the generated log files.
- Collect information that can help improve the UI.
- Bring findings to the Friday meeting for group discussion.

6. Information to Look For in Logs

Log Information	Why It Matters for the UI
Clone information	Can show clone type, file name, line numbers, and detected ranges to the user.
Validation results	Can show whether usefulness, compilation, and testing passed or failed.
Confidence score	Can help explain how confident the system is in the detection or suggestion.
Errors and warnings	Can support clearer error messages, such as compilation errors, failed tests, or unclear AI outputs.
UI-relevant details	Can help decide what information should be visible in the prototype and what should be hidden in details.

7. Issues Encountered

While testing and exploring the tool, the team identified issues that may affect setup, testing, and future UI design decisions.

Issue	Description
API key format	Google changed the format for API keys, so the program currently may not accept a newer Google API key format. The UI or settings validation may need to support the new prefix.
IntelliJ setup and behavior	IntelliJ was sometimes slow or confusing to use. It also tried to correct code that was already syntactically correct, which made testing less smooth.

8. Confidence Score Work

Dhir and Juliana are working on the confidence score section. The goal is to help users understand how safe and reliable an AI-generated Extract Method refactoring is before applying it.

One proposed direction is to calculate the score using signals from the usefulness, compilation, and testing agents, plus how many attempts the system needed before producing a successful refactoring.

- Usefulness result: did the suggestion actually remove duplicated code correctly?
- Compilation result: does the refactored Java code compile?
- Testing result: do tests pass after the refactoring?
- Number of refactoring attempts: did the system need many retries before success?
- LLM or detection confidence: how confident was the system in the initial detection or suggestion?

9. UI Improvement Discussion

On Friday, the team planned to gather information from testing and share ideas for improving the current UI design. The goal is to make the prototype clearer, more trustworthy, and more similar to the IntelliJ environment.

Idea	Purpose
Loading screen	Show that refactoring is in progress and estimate how long it may take.
Error pop-ups	Show clear messages for different errors, similar to the errors found while testing the tool.
Refactoring attempts	Show how many refactoring attempts have been made while the tool is loading or retrying.
Confidence score	Display a clear trust score so developers know how safe the suggestion is.
API key format update	Allow settings validation to accept the newer Google API key format, not only the older AlzaSy format.
IntelliJ-style design	Explore IntelliJ design patterns so the prototype feels more familiar to developers using the IDE.

10. Tasks and Next Steps

- Brainstorm how to calculate the confidence score using usefulness, compilation, testing, and retry information.
- Run and test the AntiCopyPaster tool and continue reporting issues.
- Add team members work and weekly files to the Summer Research Website.
- Upload resources used during the week to OneDrive.
- Watch the setup_gemini video to learn how to generate a free Gemini API key.
- Watch the tool demo video to better understand how the tool is used.
- Explore IntelliJ design so the UI prototype can feel closer to the real IDE.
- Continue improving the current prototype and include useful findings from logs and testing.

11. Conclusion

Overall, Week 2 focused on testing the tool, identifying setup and API key issues, planning how logs can improve the UI, and organizing the project through GitHub, OneDrive, SPARK, and the hosted progress website. The next focus is to keep testing, finish the confidence score section, improve the IntelliJ-style UI prototype, and keep the weekly documentation organized.